

SKF®

Load Path Patterns in Bearings

Presented at the bearing damage course, WE 204

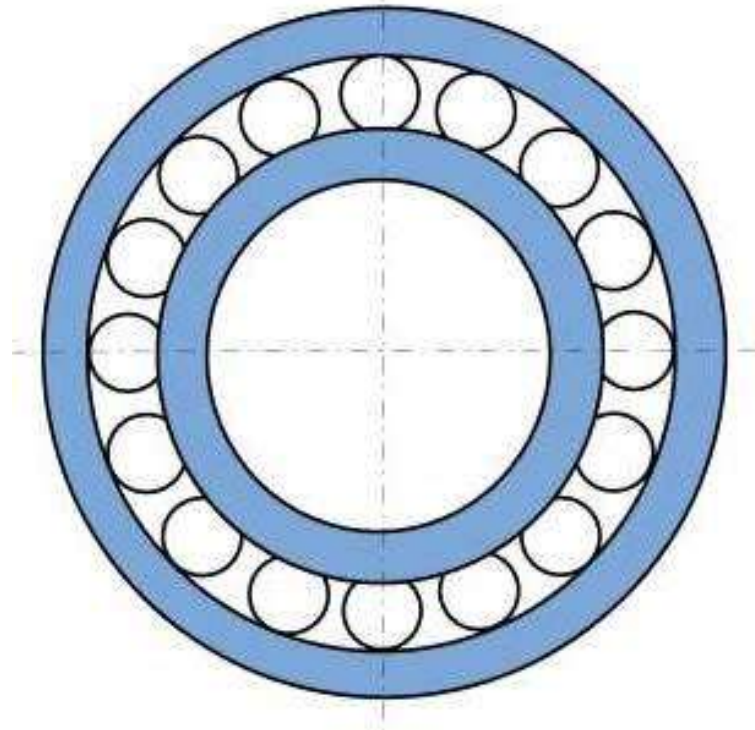
Presented by xxx

2025-05-29

1

Normal load patterns

New bearings



Change in appearance...

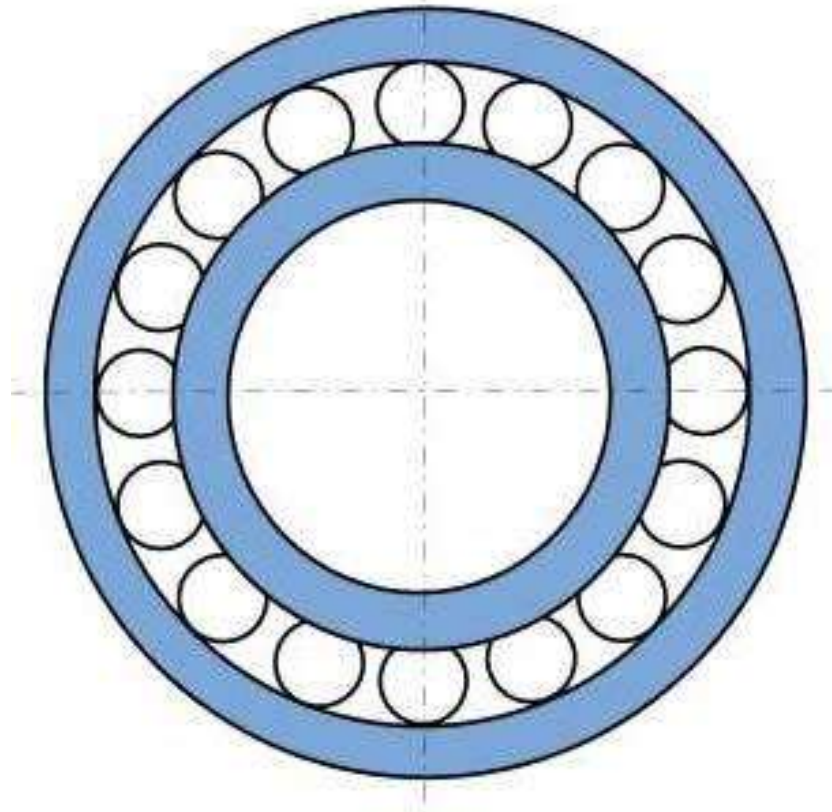
Example : electrical motor



Change in appearance



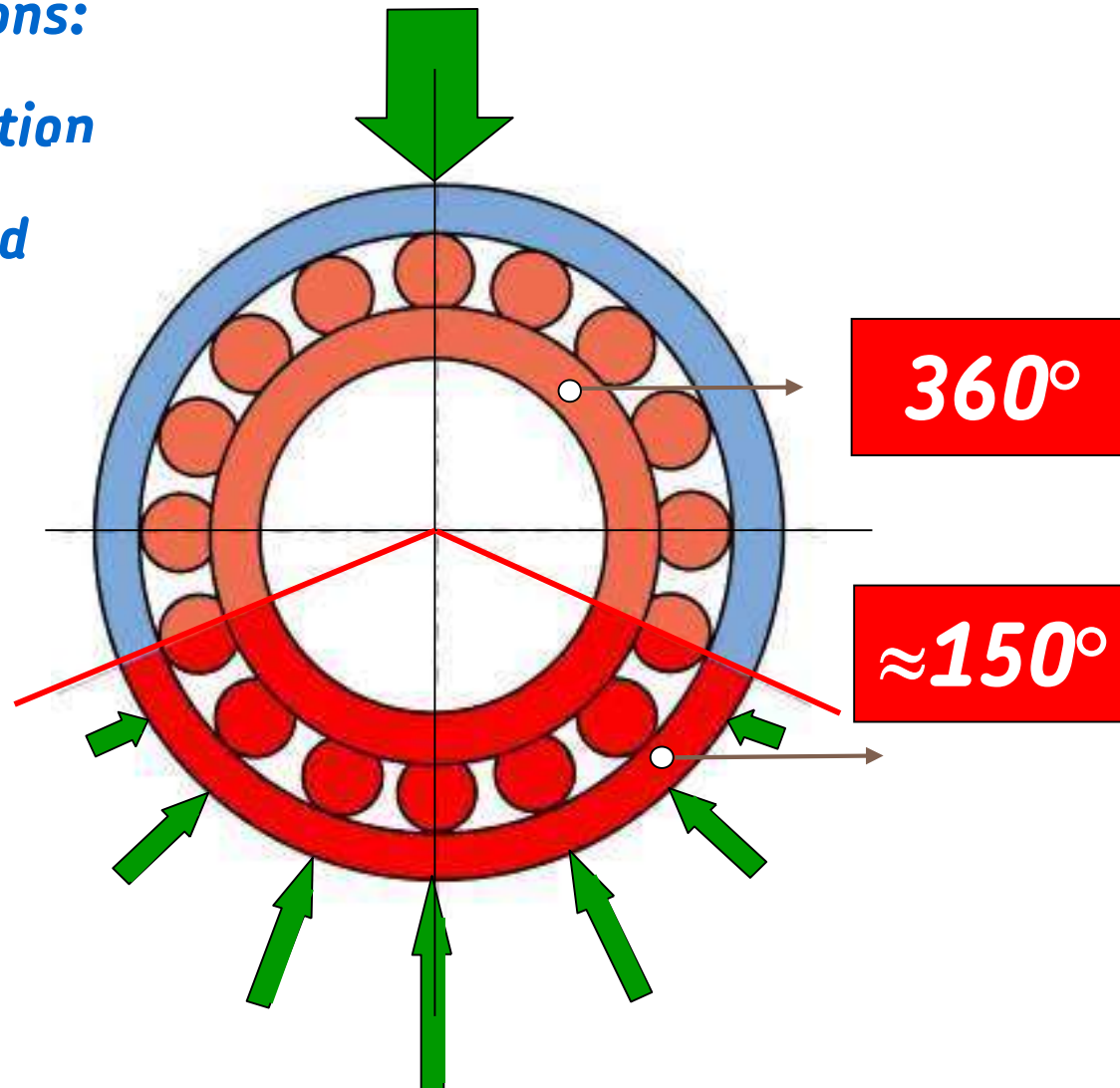
Load path pattern – normal radial loads



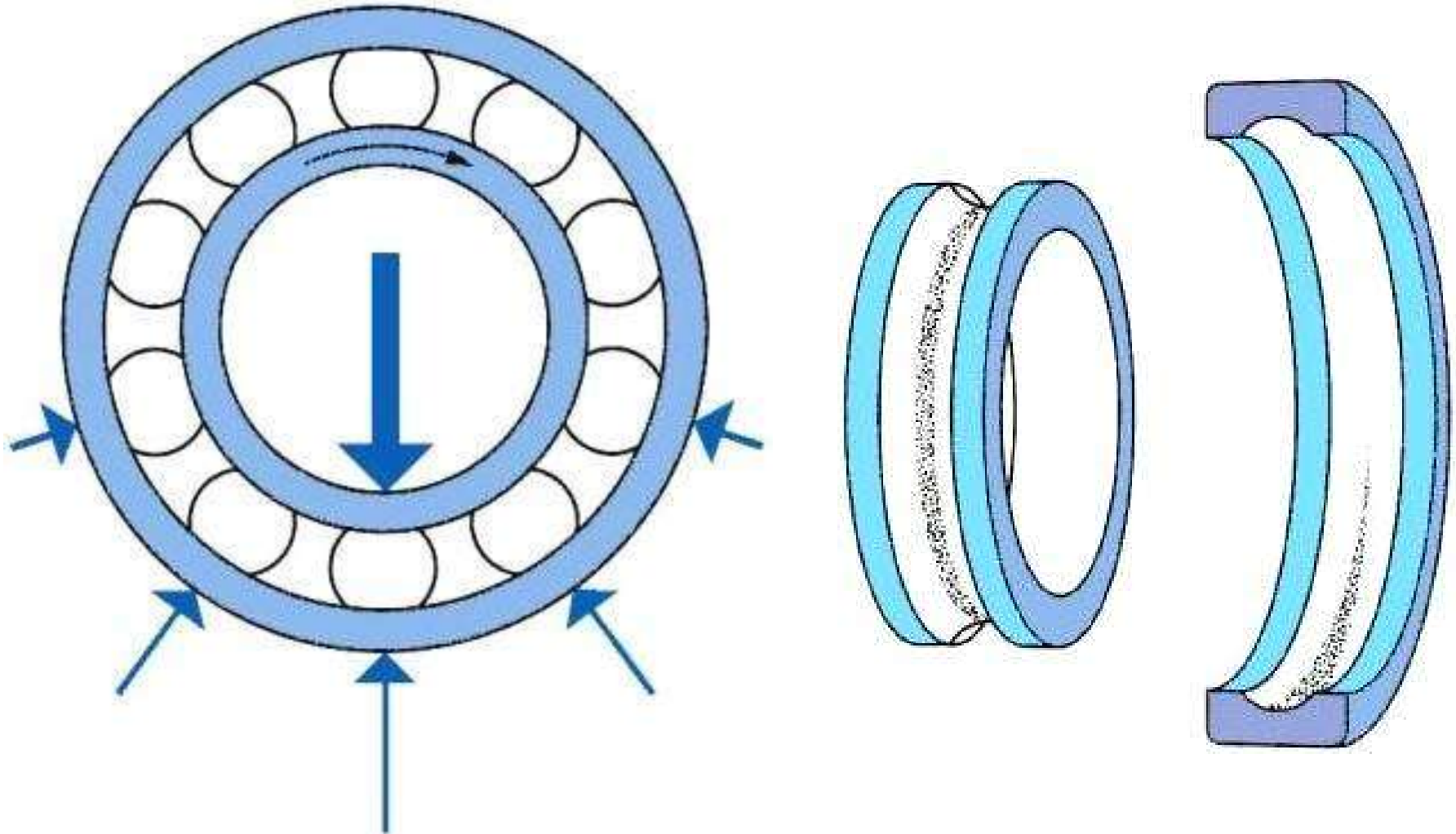
Load distribution in a bearing

Operating conditions:

- *Inner ring rotation*
- *Pure radial load*

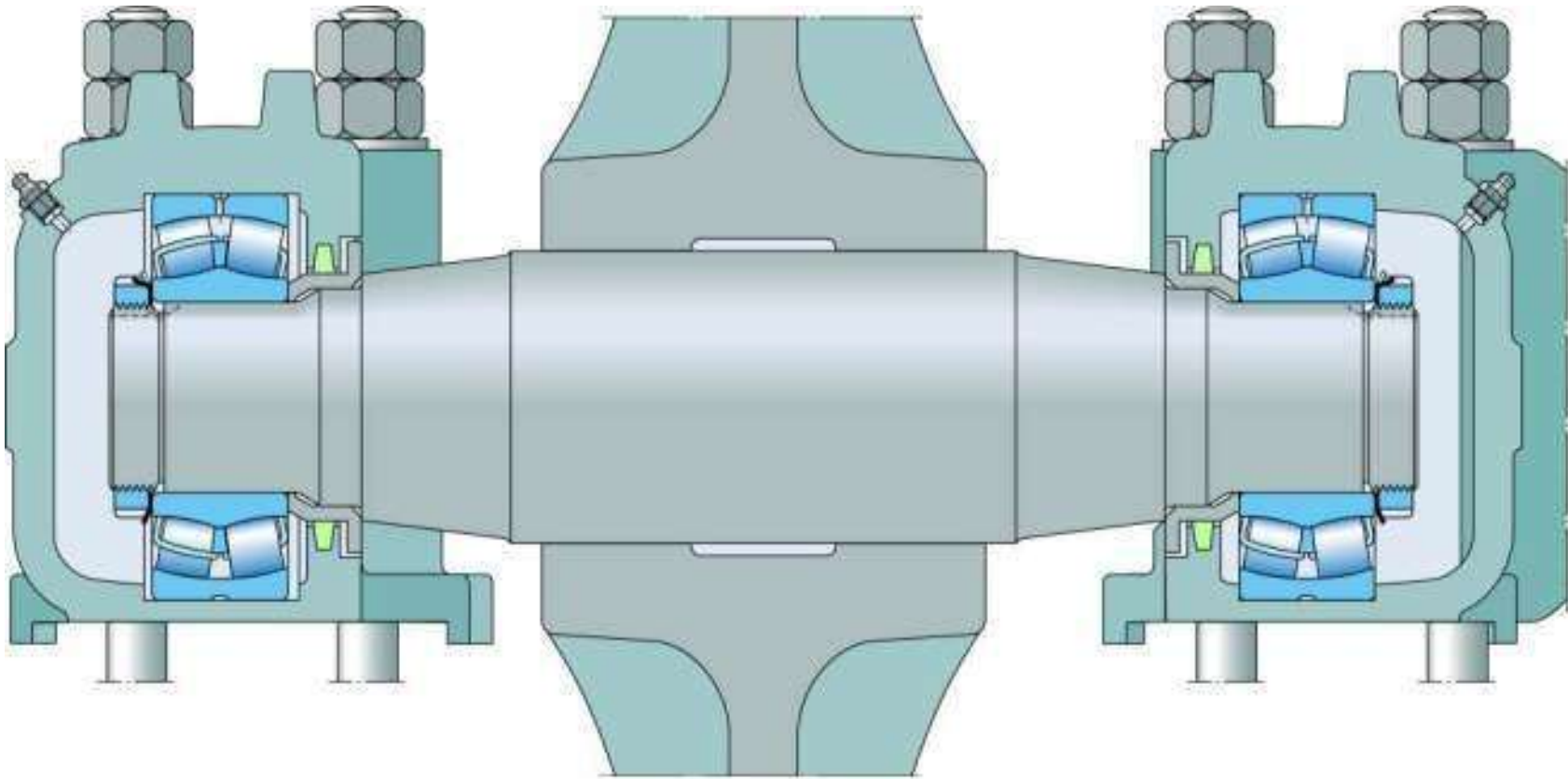


Inner ring rotation – stationary radial load

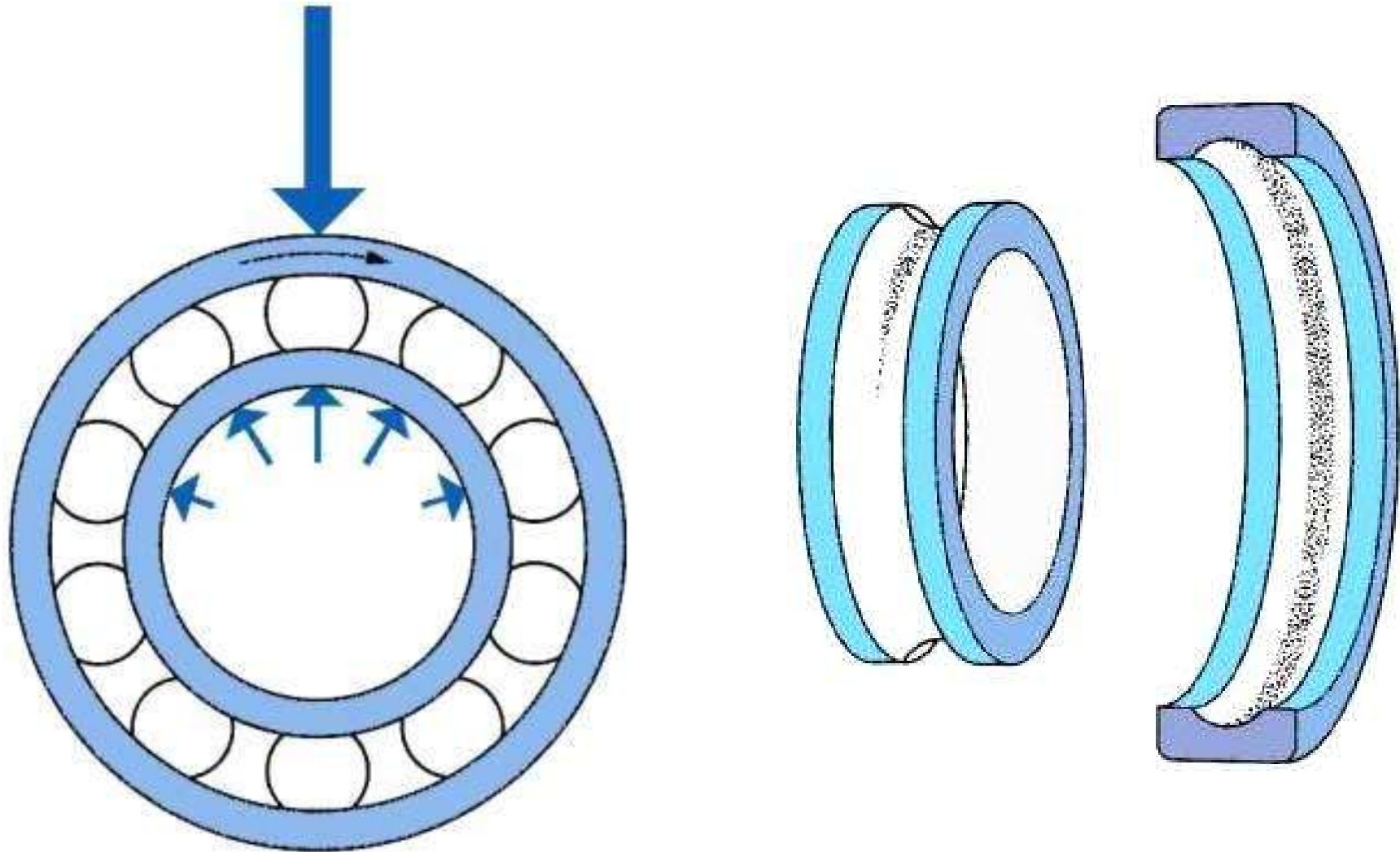


Inner ring rotation – stationary load

Example: Sheave wheel supported in two housings

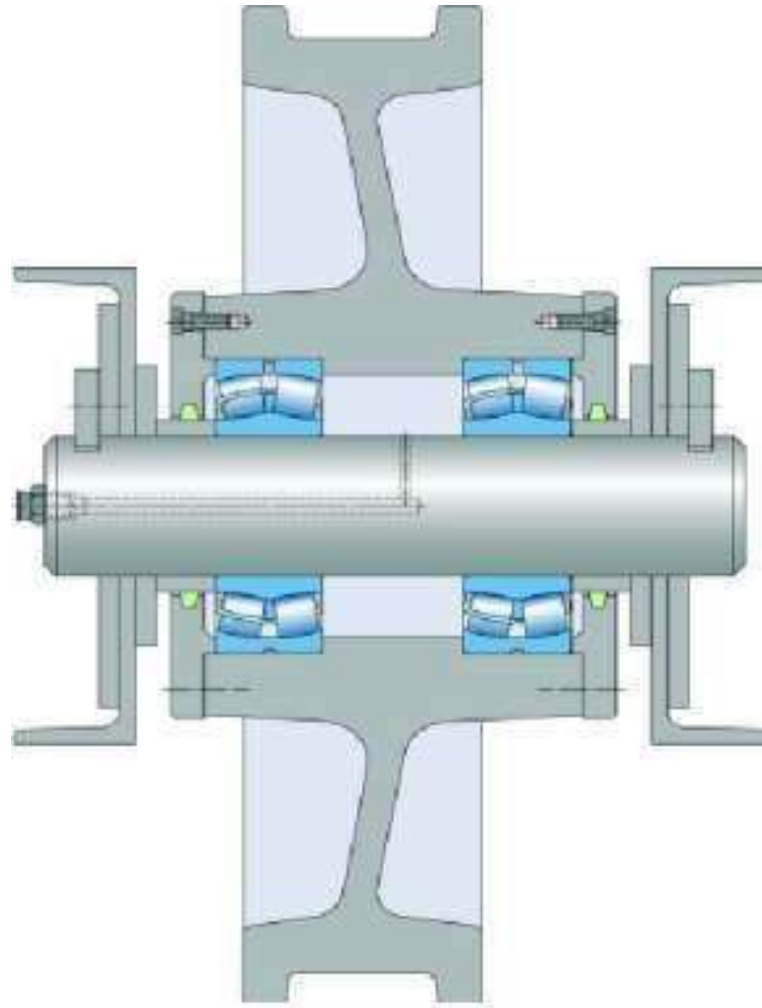


Outer ring rotation – stationary radial load...

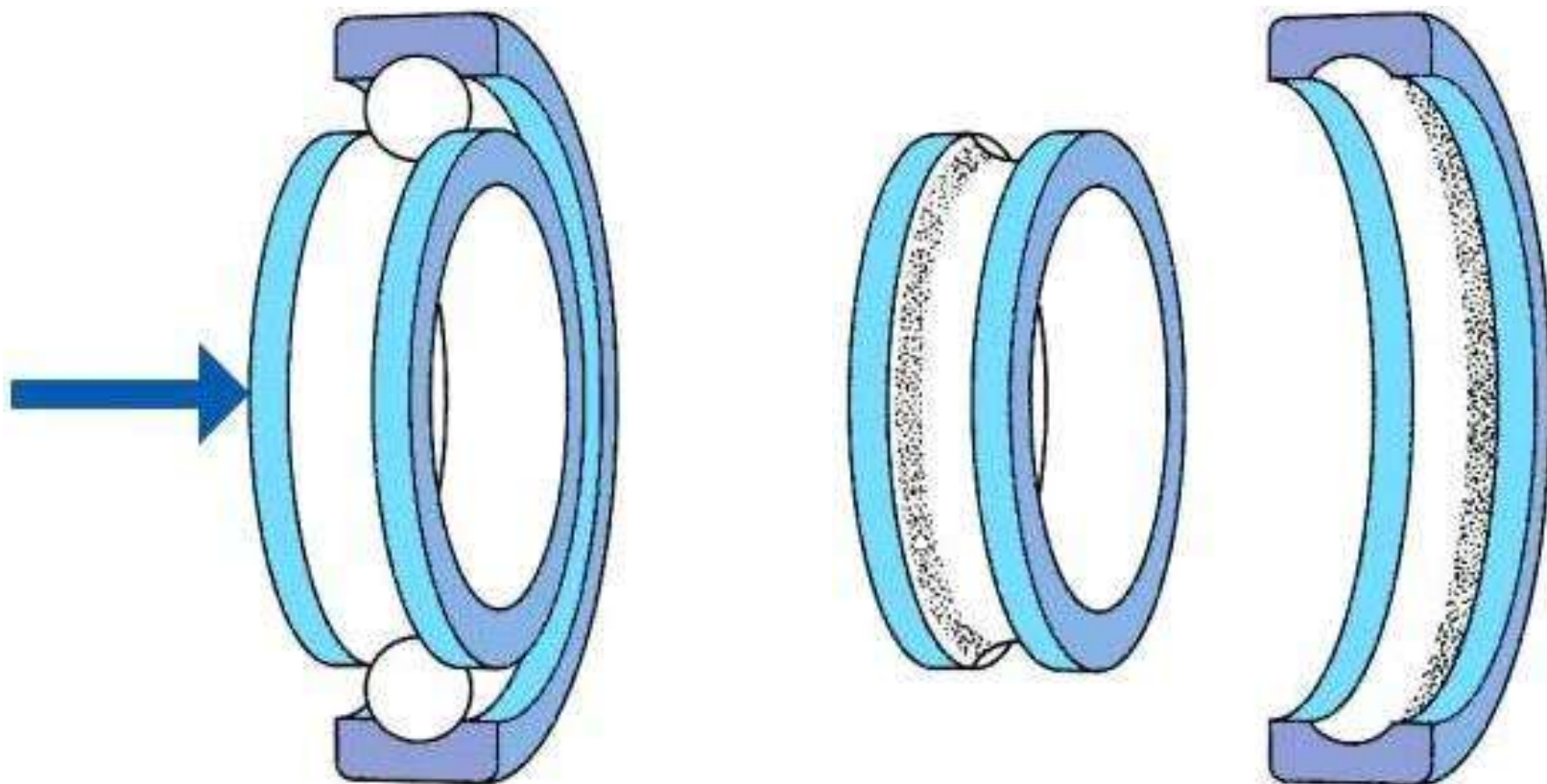


Outer ring rotation – stationary radial load

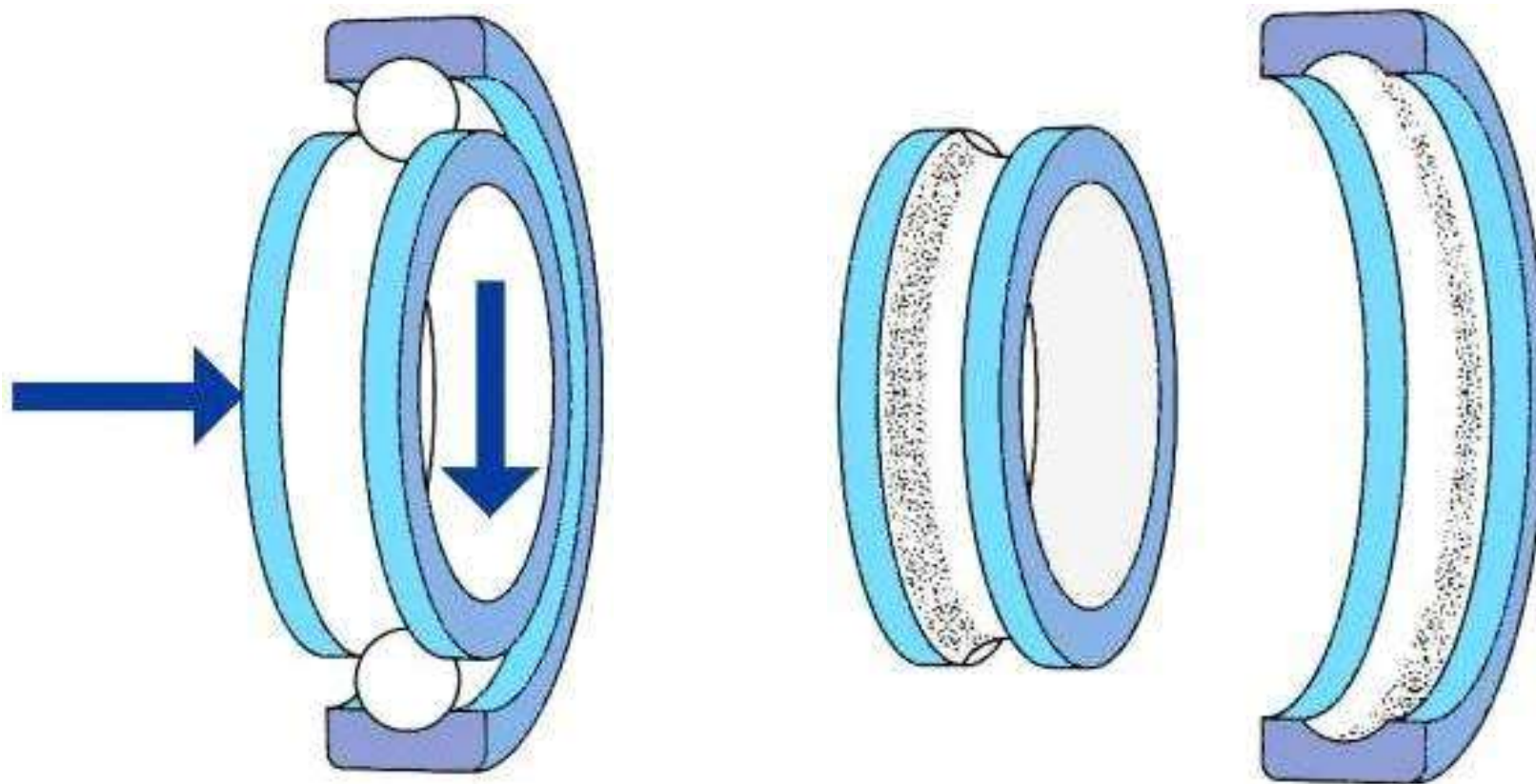
Example: Crane wheel



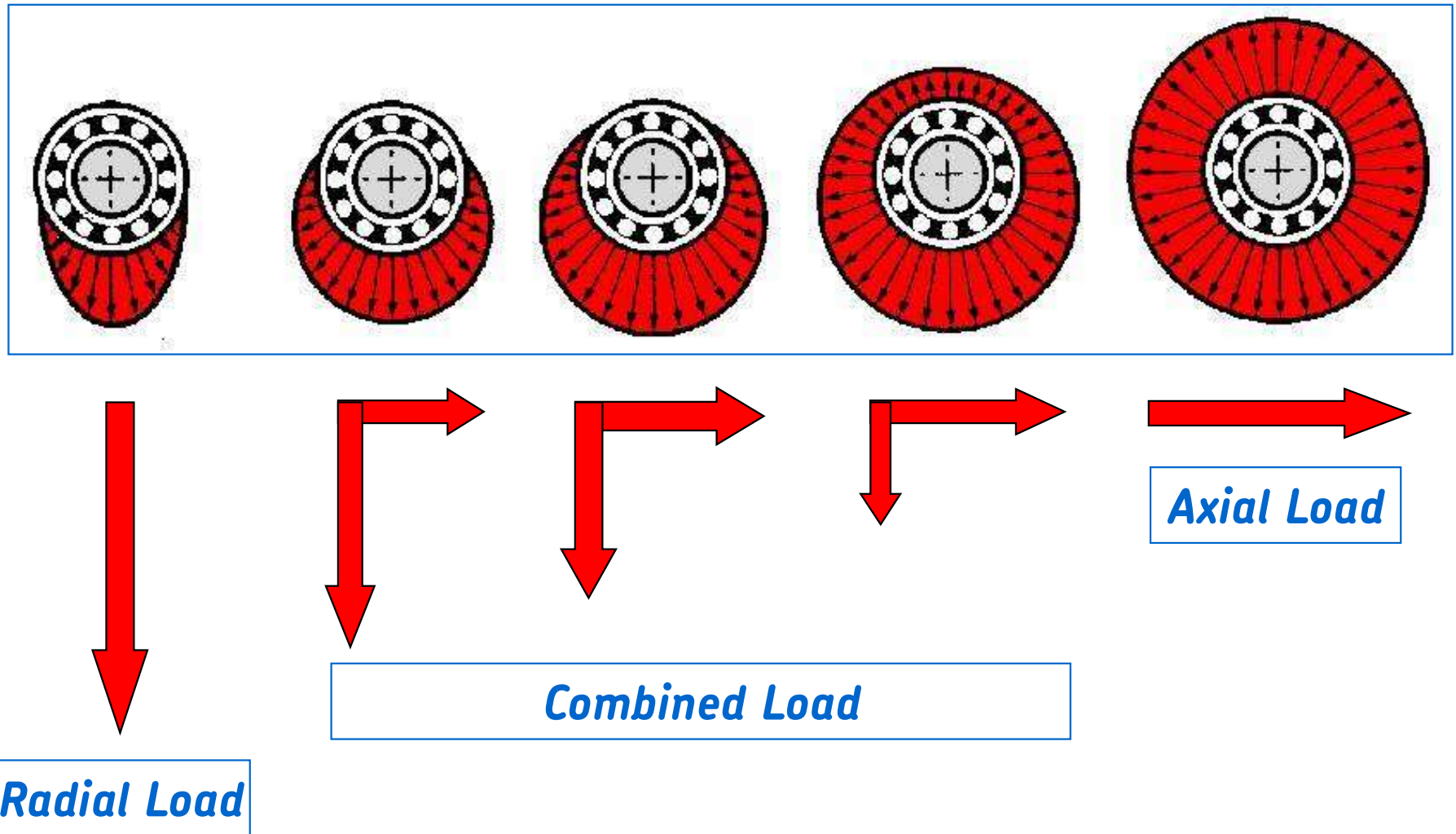
Stationary axial load



Combined loads – rotating inner ring...



Combined loads – rotating inner ring



2

Abnormal load patterns

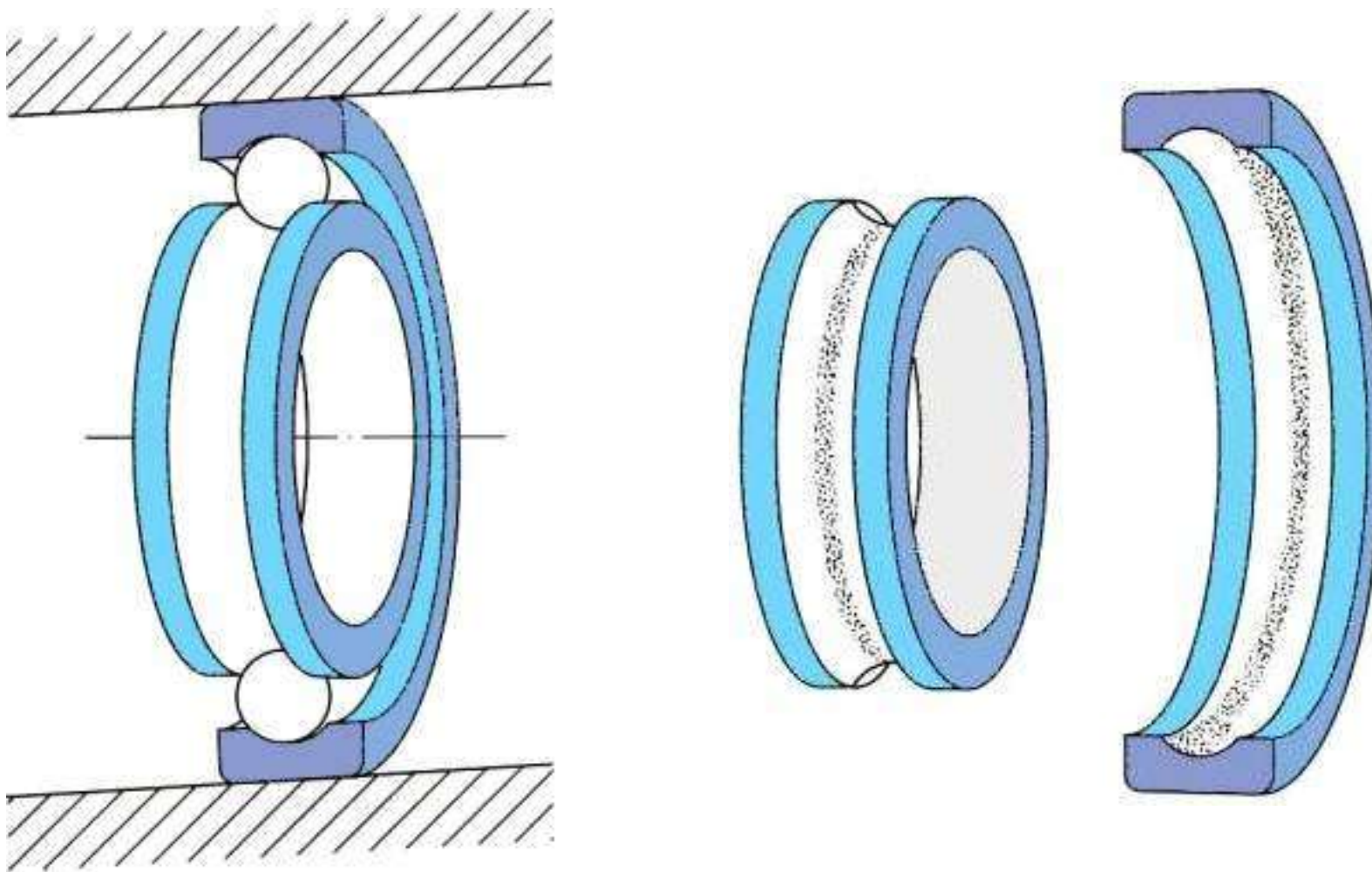
Abnormal load patterns

Any load conditions, not according to the expected ones.

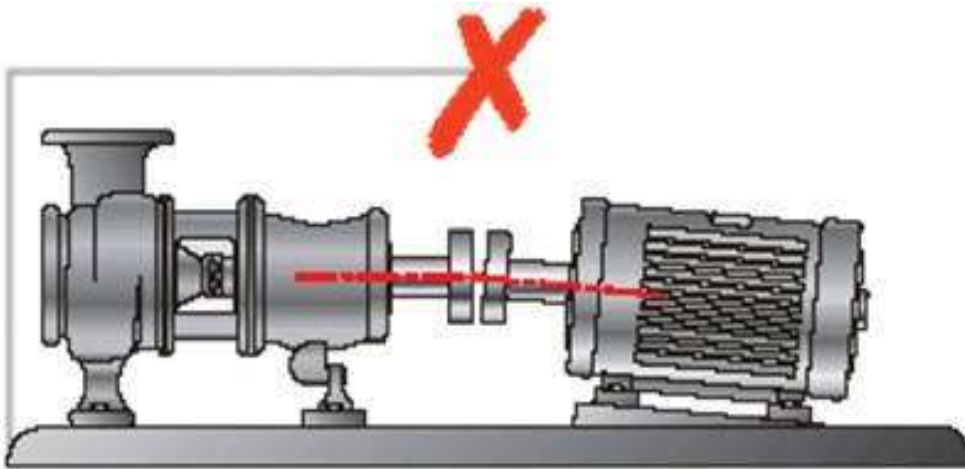
Naming a few:

- *Too tight fits*
- *Oval compression of a ring*
- *Initial misalignment*
- *Eccentricity*
- *Lubrication conditions*
- *Inadequate or wrong fitting...*

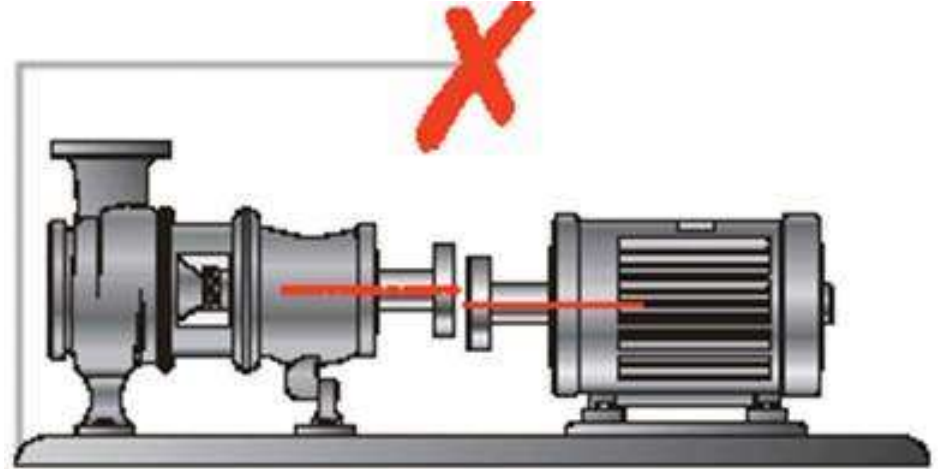
Outer ring misaligned, rotating inner ring



Initial misalignment



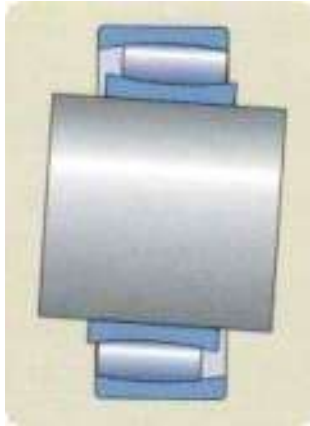
Angular Misalignment



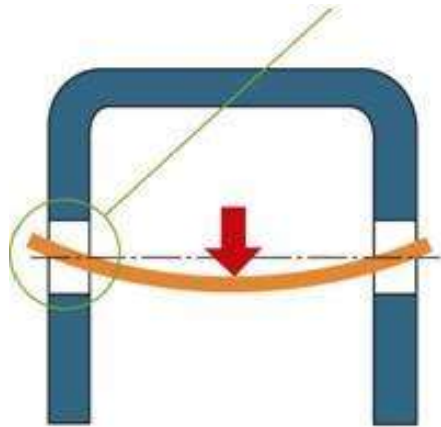
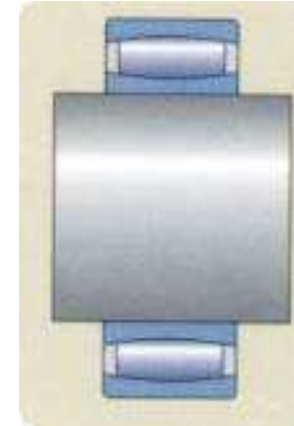
Parallel Misalignment

Misalignment

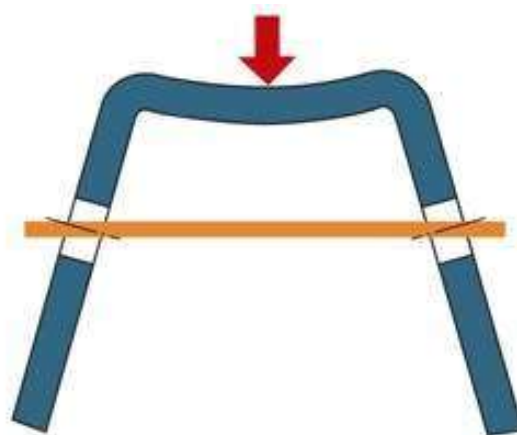
*Inner ring
misaligned to
the outer
ring*



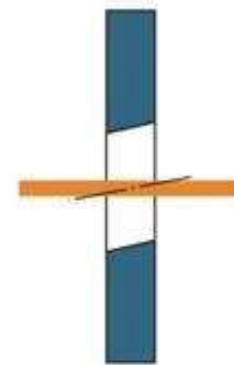
*Inner ring and
outer ring in
alignment*



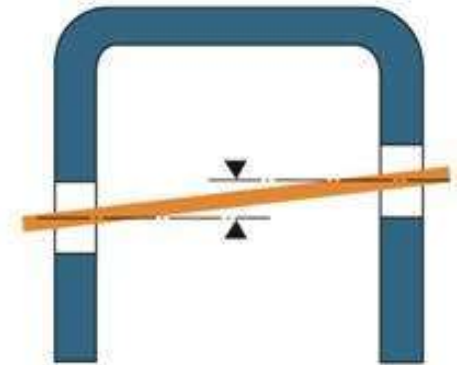
Deflected Shaft



Deflected Housing

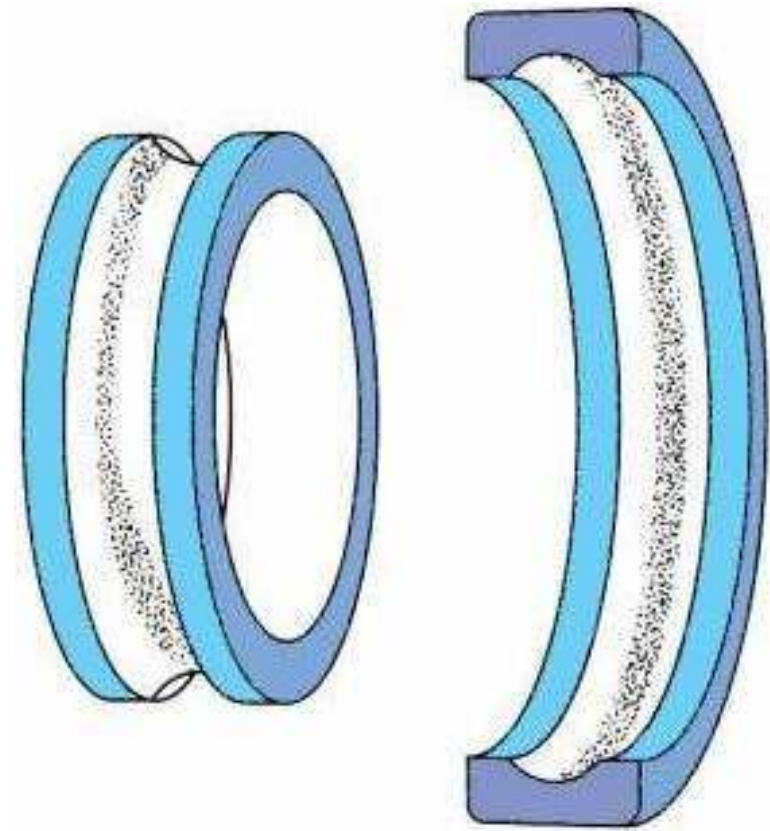
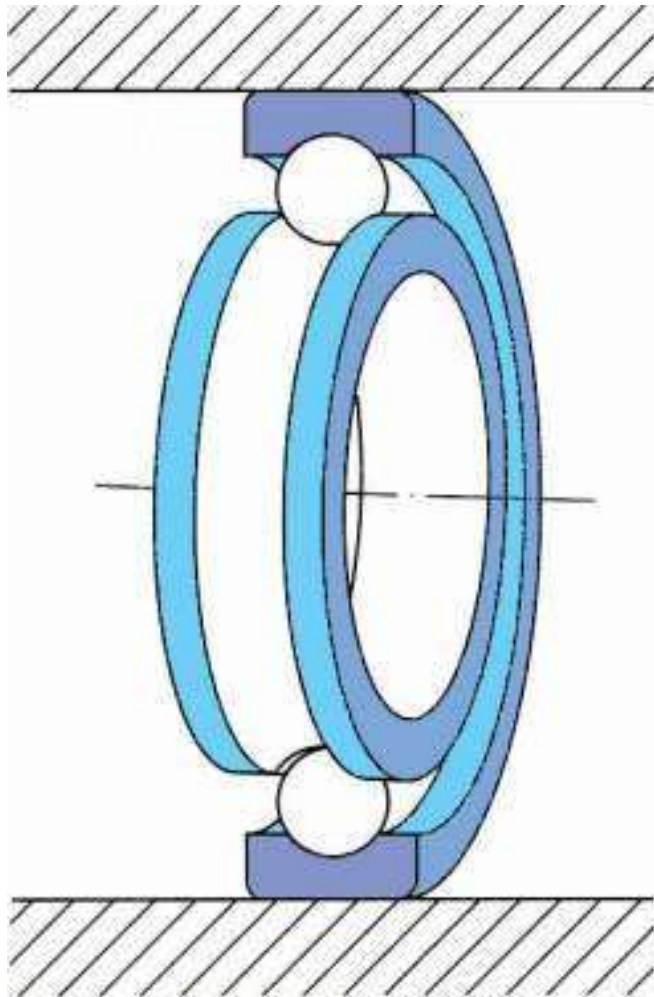


***Seating Out-
of-Square***

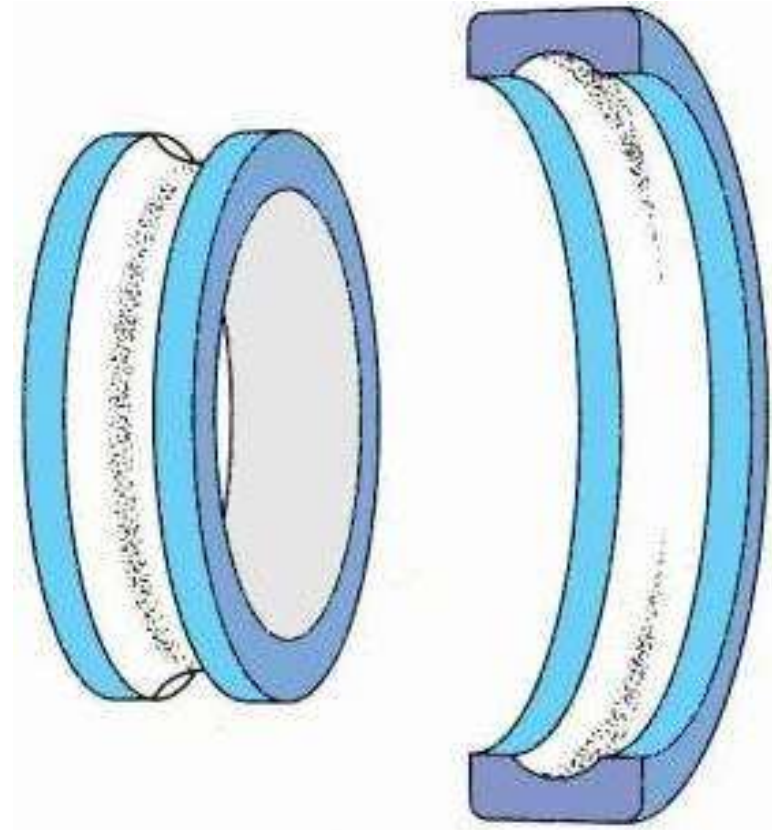
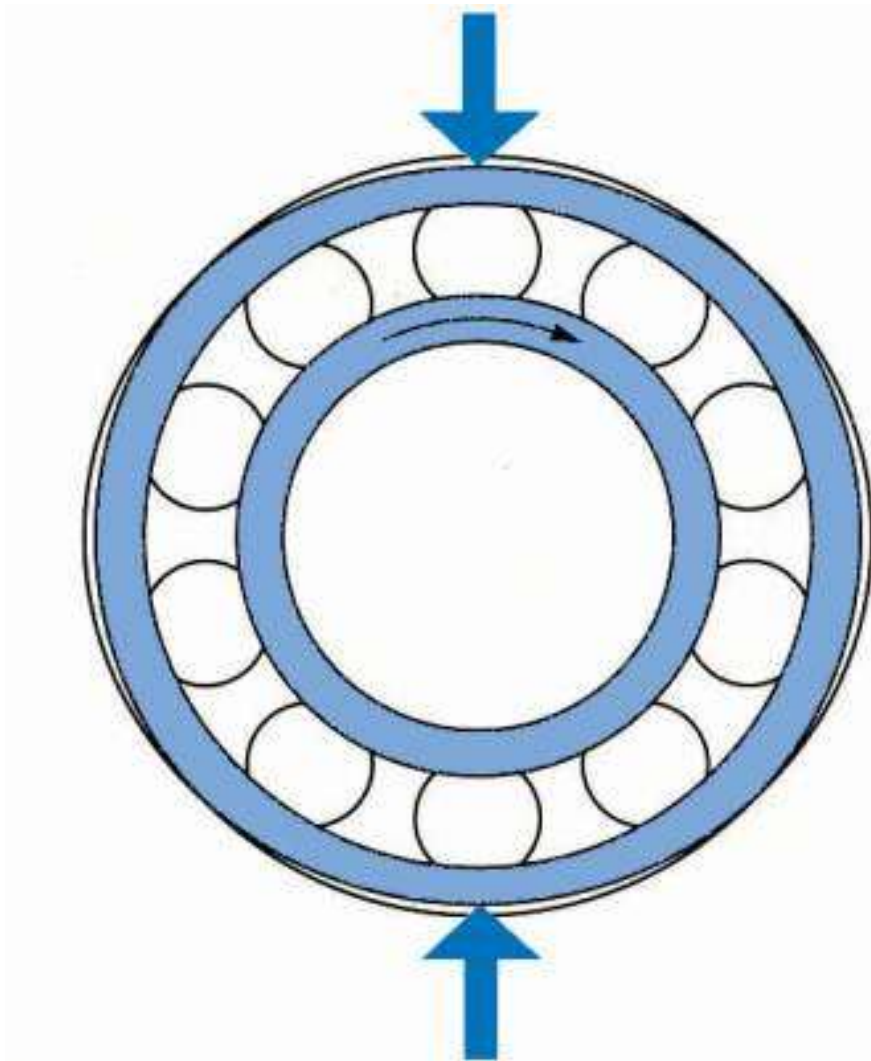


Seatings Offset

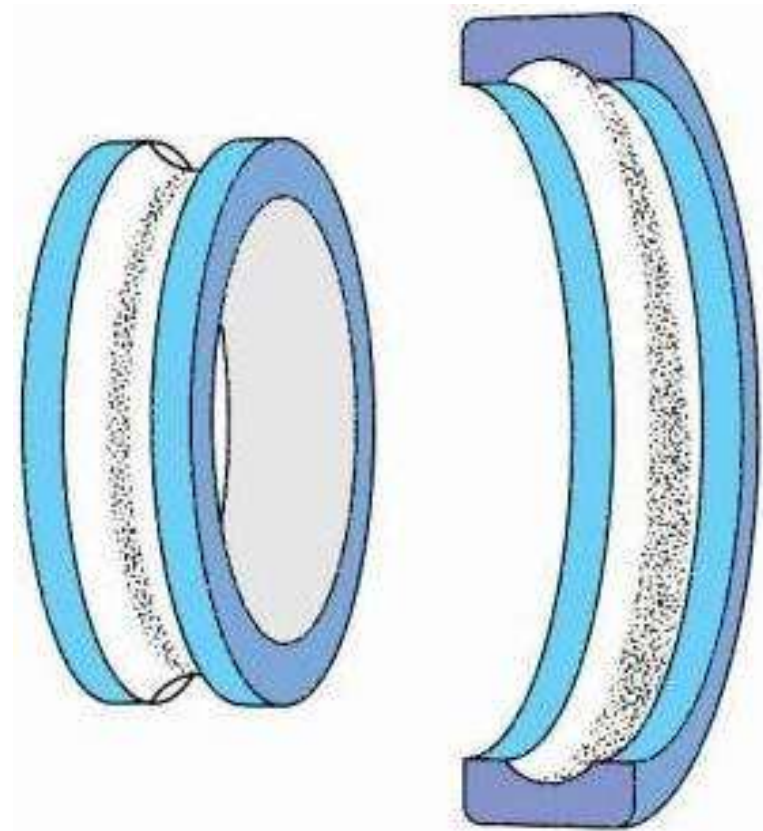
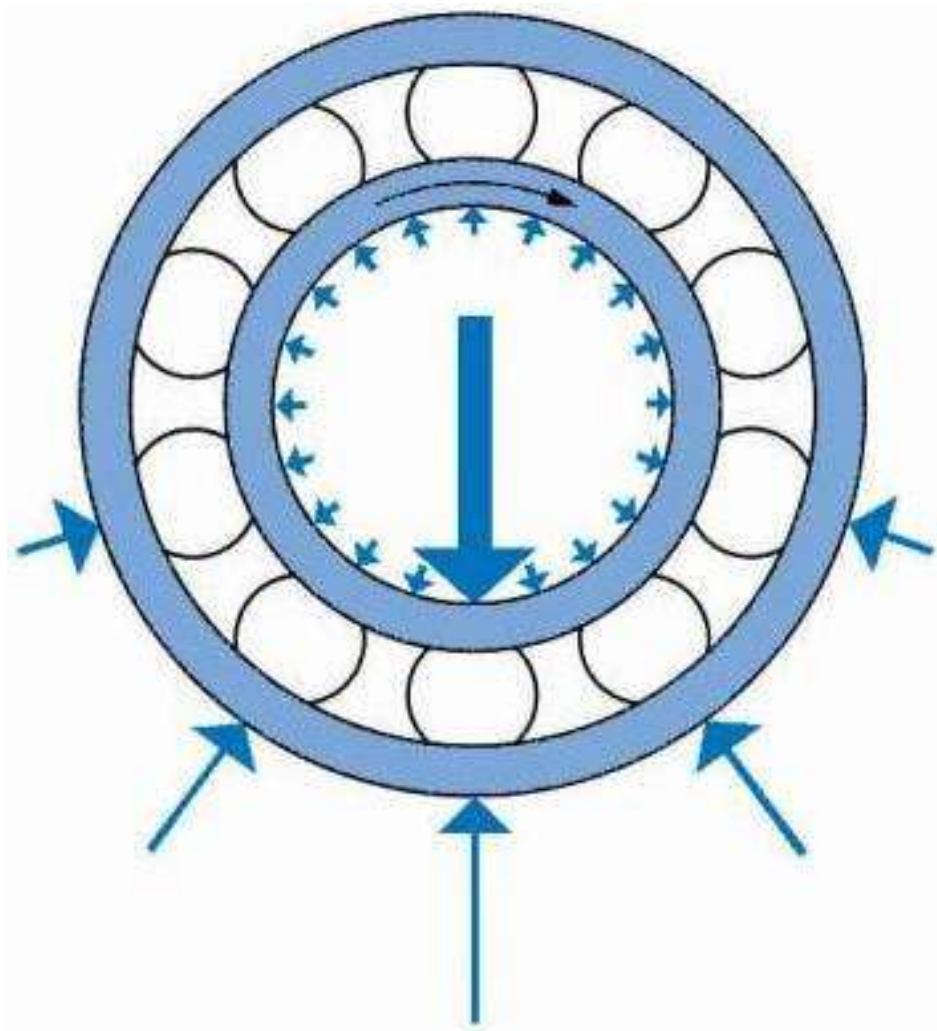
Inner ring misaligned, rotating outer ring



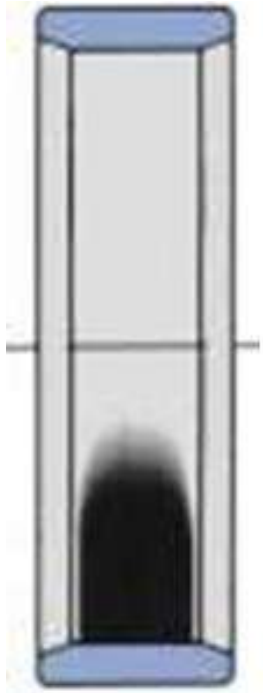
Oval compression of outer ring



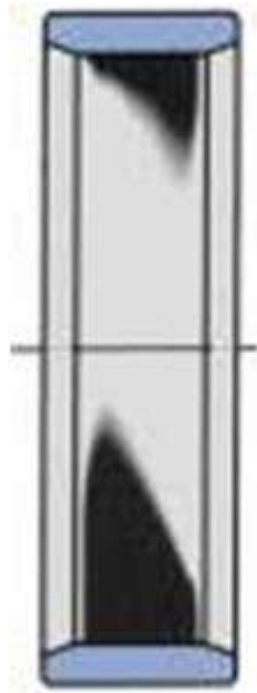
Preload – negative radial clearance



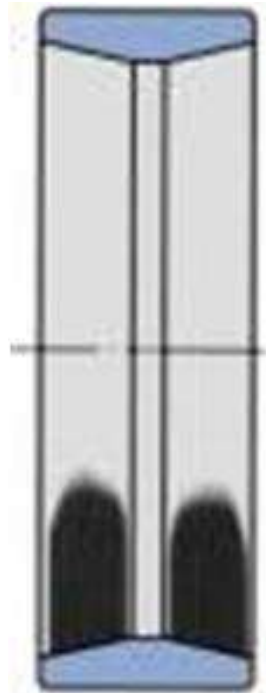
Roller bearings – patterns



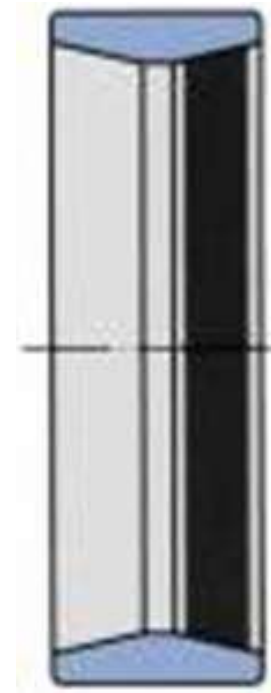
A



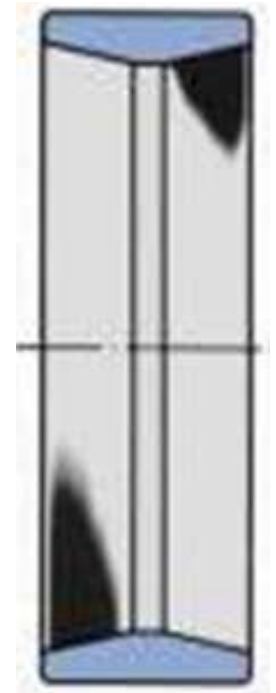
B



C



D



E

Load path patterns – last word

In operation, path patterns appear in areas where load is transmitted. Some of them are straightforward; others more complex. Some examples:

- *Contaminants have come in and have been over-rolled, or*
- *The lubricant did not have the right viscosity, or*
- *The fitting was not done correctly, or*
- *The tolerances for the bearing seats were not correct, or*
- *Temperature increase, ring growth, loss of clearance, etc.*

They all result in some “typical” signs or patterns.

There are a large number of them.

There is need to understand and to classify the different types of damage.

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